Unit 2 Pre-Test

Exponent Rules & Properties

Standard assessed: NC.8.EE.1

ANSWER KEY

Name: _____ **Date:** _____ **Period:** _____

SCORE **3.0** **Basic Understanding**

NC.8.EE.1 Develop and apply the properties of integer exponents to generate equivalent numerical expressions.

• *I can apply the properties of integer exponents to simplify expressions with numerical bases.*

Match each vocabulary term to its definition, then simplify each expression.

D base ✓	**A.** The numerical factor multiplied by the variable(s) in a term. *Example: the 7 in $7y^2$*
B exponent ✓	**B.** Tells how many times the base is used as a factor. *Example: the 3 in $6^3 = 6 \cdot 6 \cdot 6$*
A coefficient ✓	**C.** Terms with identical variable parts (same variables to the same powers). *Example: $4a^2b$ and $-9a^2b$*
F monomial ✓	**D.** A number or variable used as a repeated factor in a power. *Example: the y in $y^4 = y \cdot y \cdot y \cdot y$*
C like terms ✓	**E.** An expression that uses a root symbol, such as a square root (√). *Example: $\sqrt{36} = 6$*
E radical ✓	**F.** A single term: a number, a variable, or a product of numbers and variables. *Example: 8, m, or $-3p^2q$ (but not m + 8)*

Parts of a monomial — reference

$$5x^3$$

coefficient	**base**	**exponent**
the number factor	the repeated factor	how many times the base is a factor

A **monomial** *is a single term — a number, a variable, or a product such as the one above.*

1.

Simplify:

$4^3 \cdot 4^2$

answer: 4^5

2.

Simplify:

$(5^2)^3$

answer: 5^6

3.

Simplify:

$$\frac{6^7}{6^4}$$

answer: 6^3

4.

Simplify:

$2^4 \cdot 2^4$

answer: 2^8

5.

Simplify:

$(10^3)^2$

answer: 10^6

6.

Simplify:

$$\frac{9^5}{9^2}$$

answer: 9^3

SCORE 4.0 Target Content

NC.8.EE.1 Develop and apply the properties of integer exponents to generate equivalent numerical expressions.

• *I can multiply and divide monomials using the properties of exponents.*

Multiply, divide, or raise to a power. Simplify each monomial expression.

7.

Multiply:

$2x^3 \cdot 5x^4$

answer: $10x^7$

8.

Multiply:

$(4a^2b)(3a^3b^5)$

answer: $12a^5b^6$

9.

Divide:

$$\frac{12x^6}{3x^2}$$

answer: $4x^4$

10.

Divide:

$$\frac{20a^7b^4}{5a^3b^2}$$

answer: $4a^4b^2$

11.

Simplify:

$(2x^3)^4$

answer: $16x^{12}$

12.

Simplify:

$(3x^2y^4)^3$

answer: $27x^6y^{12}$

SCORE **5.0** **Advanced Content**

NC.8.EE.1 Develop and apply the properties of integer exponents to generate equivalent numerical expressions.

• *I can simplify complex, multi-step monomial expressions, including zero and negative exponents.*

Simplify using several exponent properties. Write answers with positive exponents.

13.

Simplify:

$$\frac{(2x^3y^4)(5x^2y)}{10x^5y^5}$$

answer: 1

14.

Simplify:

$$\frac{4a^5b^3}{4a^5b^7}$$

answer: $\frac{1}{b^4}$

15.

Simplify:

$$(2x^{-2})^3$$

answer: $\frac{8}{x^6}$

16.

Simplify:

$$(4x^3y^{-2})(2x^{-5}y^4)$$

answer: $\frac{8y^2}{x^2}$

17.

Simplify:

$$\frac{(3x^2y^4)^2}{9x^5y^6}$$

answer: $\frac{y^2}{x}$

18.

Simplify:

$$\frac{(6x^5y^2)(2xy^3)}{3x^4y^7}$$

answer: $\frac{4x^2}{y^2}$